SE LAB 1- Scenario 29

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Req Table

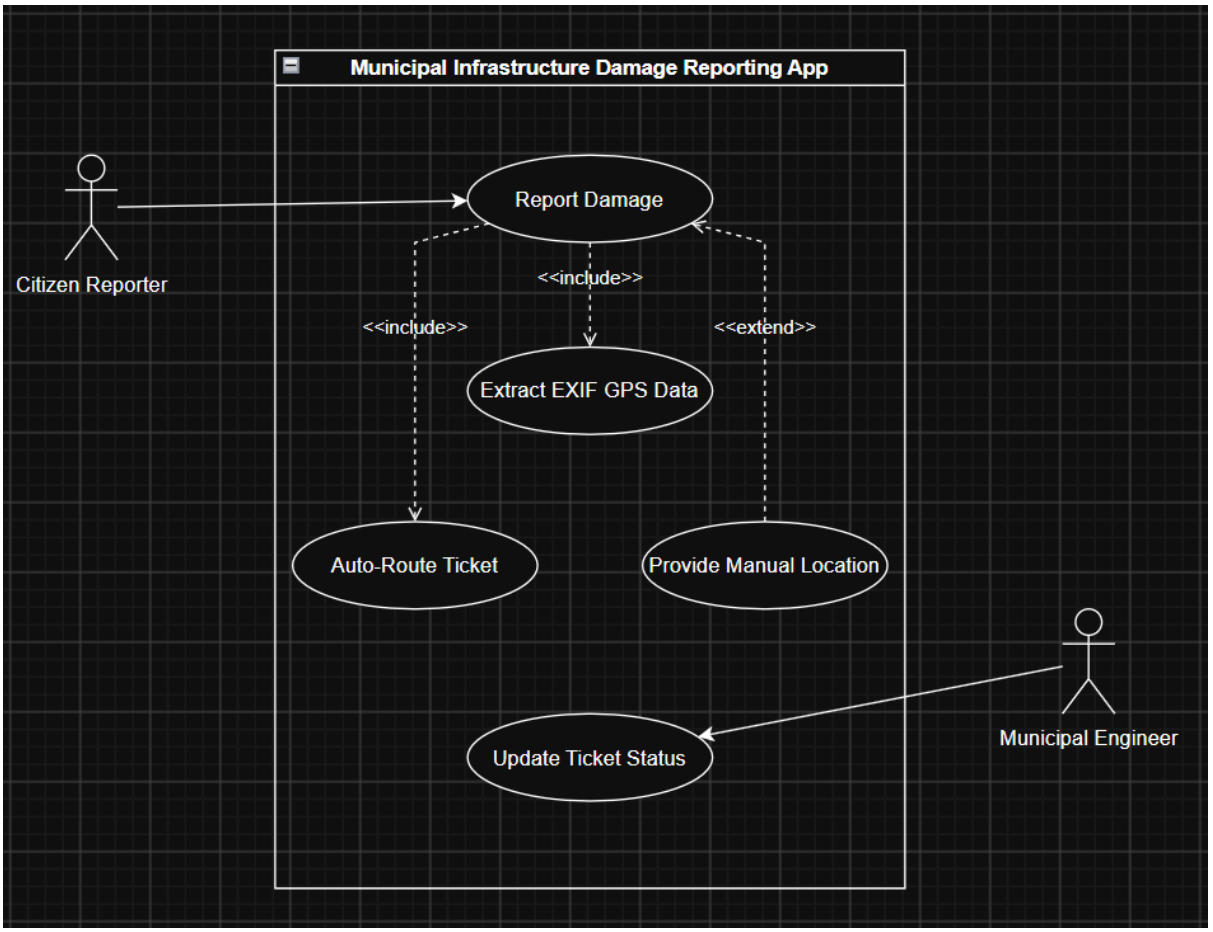
ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall extract EXIF GPS metadata from uploaded damage photos to automatically pinpoint incident location on the municipal GIS map.	High	Pass: Pothole report routed to Road Infrastructure division with exact GPS coordinates.	Ensures accurate location tracking without relying on manual user entry, facilitating faster repairs.
				Fail: Report submitted without location data.	
FR-002	Functional	The system shall automatically dispatch submitted tickets to the relevant municipal engineering departments based on the selected issue category (e.g., roads, water, electrical).	High	Pass: Water pipeline leak report appears only on the Water Department's dashboard.	Automates the triage process, saving manual dispatcher time and ensuring quick resolutions.
				Fail: Ticket remains unassigned or is routed to the wrong department.	
FR-003	Functional	The portal shall allow the Citizen Reporter to manually drop a pin on an integrated map if the	Medium	Pass: User successfully submits a report with a manually	Acts as a fallback mechanism for users taking photos without

		uploaded photo lacks EXIF GPS metadata.		selected map location.	location services enabled.
				Fail: User is blocked from submitting a report if EXIF data is missing.	
FR-004	Functional	The system shall allow the Municipal Engineer to update the status of an assigned ticket (e.g., "Received", "In Progress", "Resolved").	High	Pass: Engineer changes status to "Resolved" and it reflects immediately on the system.	Keeps the system updated with real-time progress on infrastructure repairs.
				Fail: Engineer is unable to modify the ticket status.	
FR-005	Functional	The system shall automatically notify the Citizen Reporter via in-app notification when the status of their reported ticket changes.	Low	Pass: Citizen receives a notification when their pothole report is marked "Resolved".	Improves transparency and keeps citizens engaged in the municipal maintenance process.
				Fail: Status changes, but no notification is triggered.	
NFR-001	Performance & Security	The reporting portal shall support offline report drafting and auto-sync when mobile network connectivity resumes.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Essential for users reporting issues in areas with poor cellular coverage (e.g., severe weather damage).
NFR-002	Usability	The system's user interface shall be accessible and allow a Citizen Reporter to submit a damage report in fewer than 3	Medium	Pass: A new user can complete a report submission	Ensures the application is user-friendly and encourages civic participation

		screens/clicks from the home screen.		within 3 distinct navigation steps.	without a steep learning curve.
				Fail: Submission requires more than 3 steps.	

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NFR-002	Usability	The system's user interface shall be accessible and allow a Citizen Reporter to submit a damage report in fewer than 3 screens/clicks from the home screen.	Mediu m	Pass: A new user can complete a report submission within 3 distinct navigation steps.	Ensures the application is user-friendly and encourages civic participation without a
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UML Diagram



Main Success Scenarios and Alternate FLow

Preconditions:

- The Citizen Reporter has downloaded the app and is logged in.
- The device has a working camera and location services enabled (or offline drafting is active).

Postconditions:

- A unique ticket is generated and logged in the system.
- The ticket is automatically routed to the correct municipal department based on category.

Main Success Scenario:

1. The Citizen Reporter opens the app and selects "Report New Damage".
2. The Citizen Reporter selects the category of damage (e.g., Road Pothole, Broken Streetlight, Pipeline Leak).
3. The Citizen Reporter takes or uploads a geotagged photo of the damage.

4. The System automatically extracts EXIF GPS metadata from the photo to pinpoint the location (<> *Extract GPS Data*).
5. The Citizen Reporter adds an optional text description and clicks "Submit".
6. The System generates a ticket, dispatches it to the relevant department, and displays a success confirmation.

Alternate Flow (1a): Missing GPS Metadata

- *Condition:* At step 4, the uploaded photo lacks EXIF GPS metadata.
- *Action:*
    1. The System alerts the user that location data is missing.
    2. The System prompts the user to manually enter the location or drop a pin on the map (<> *Provide Manual Location*).
    3. The Citizen Reporter selects the location manually.
    4. The flow resumes at Step 5 of the Main Success Scenario.